

Math726 – Final Exam

You must show all of your work to get a credit for a correct answer.

- Given $a \leq x_0 < x_1 < \cdots < x_n \leq b$, let $f(x)$ be a sufficiently smooth function and $H_{2n+1}(x)$ be its Hermite interpolating polynomial of degree $\leq 2n+1$ satisfying $H_{2n+1}(x_k) = f(x_k)$ and $H'_{2n+1}(x_k) = f'(x_k)$ for $k = 0, 1, \dots, n$. Prove that for all $x \in (a, b)$, there exists a $\eta \in (a, b)$ such that

$$f(x) - H_{2n+1}(x) = \frac{f^{(2n+2)}(\eta)}{(2n+2)!} \omega^2(x)$$

where $\omega(x) = (x - x_0)(x - x_1) \cdots (x - x_n)$.

- Compute the coefficients and weights $\{(\lambda_j, \sigma_j)\}_{j=1}^n$ of the Gaussian quadrature formulas $\mathcal{I}_n(f) = \sum_{j=1}^n \sigma_j f(\lambda_j)$ for the approximation of $\int_a^b f(x) \rho(x) dx$ where $a = -1$, $b = 1$, $n = 2$ and

$$\rho(x) = \begin{cases} 2, & 0 < x \leq 1, \\ 1, & -1 \leq x \leq 0. \end{cases}$$

- Let f be a continuous function on $[-1, 1]$ and $\mathcal{I}_n(f)$ denote the Gaussian quadrature formulas $\mathcal{I}_n(f) = \sum_{j=1}^n \sigma_j f(\lambda_j)$ for the approximation of $\mathcal{I}(f) = \int_{-1}^1 f(x) dx$. Prove that

$$\lim_{n \rightarrow +\infty} \mathcal{I}_n(f) = \mathcal{I}(f).$$

- Consider solving the initial value problem $y' = f(t, y)$, $y(t_0) = y_0$ using the one-step method

$$u_{n+1} = u_n + h[(1 - \theta)f(t_n, u_n) + \theta f(t_{n+1}, u_{n+1})].$$

where θ is constant parameter in $[0, 1]$.

- Determine the order of this method.
 - Show that this method is A-stable when $\theta \in [1/2, 1]$.
- Consider the family of linear multi-step methods

$$u_{n+1} = u_n + \frac{h}{2}[2(1 - \alpha)f_{n+1} + 3\alpha f_n - \alpha f_{n-1}]$$

where α is a real parameter.

- Analyze consistency and order of the methods as functions of α , determine the value α^* for which the resulting method has maximal order.
- Study the zero-stability of the method with $\alpha = \alpha^*$, write its characteristic polynomial $p(r; h\lambda)$ and, using MATLAB, draw its region of absolute stability in the complex plane.

6. Consider the system of ODEs

$$\mathbf{y}'(t) = \mathbf{A}\mathbf{y}(t), \quad \mathbf{y}(0) = (1, 0, -1)^T$$

where $\mathbf{y}(t) = (y_1(t), y_2(t), y_3(t))^T$ and

$$\mathbf{A} = \begin{pmatrix} -21 & 19 & -20 \\ 19 & -21 & 20 \\ 40 & -40 & -40 \end{pmatrix}.$$

The exact solution is given by

$$\begin{cases} y_1(t) = \frac{1}{2}e^{-2t} + \frac{1}{2}e^{-40t}(\cos 40t + \sin 40t) \\ y_2(t) = \frac{1}{2}e^{-2t} - \frac{1}{2}e^{-40t}(\cos 40t + \sin 40t) \\ y_3(t) = -e^{-40t}(\cos 40t - \sin 40t) \end{cases}$$

Numerically solve this ODE system for $t \in [0, 2]$ using the standard 4-th order RK method with the stepsizes i) $h = 0.08$, ii) $h = 0.04$, and iii) $h = 0.02$. Compare your simulations with the exact solution and plot the errors for each component y_i .